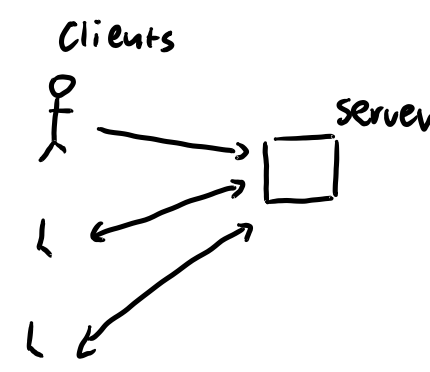


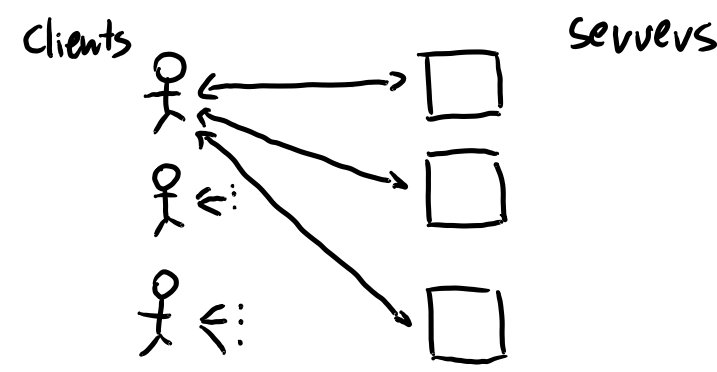
Intro

Client Server



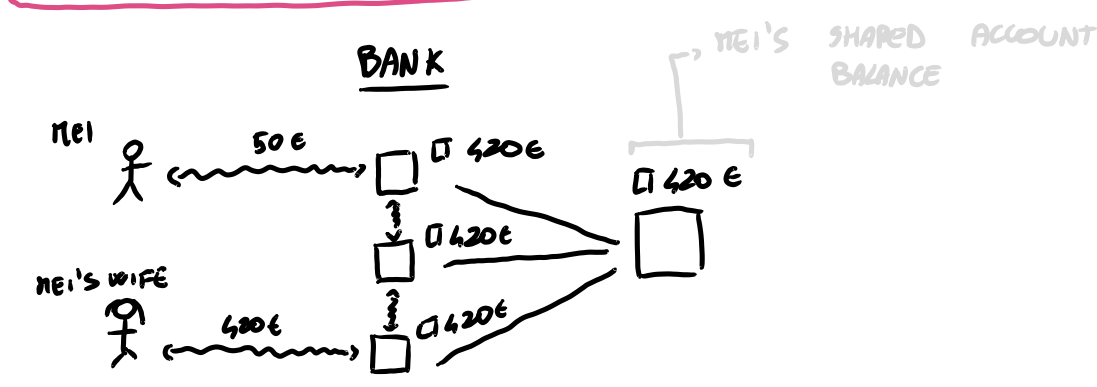
- Does not scale
- Single Point of Failure
- Geographical problems

Distributed System



- Scales
- Resilient / Fault-Tolerant
- Also distributed Geographically

Possible Problems



Order of operation is important

- We can exchange messages for coordination but they can be lost
- We ask: Can I coordinate servers with a protocol with a finite number of steps (messages)?
- ↳ Two General problem → long story short: No, we can't

Classification

→ Me's Military service example

→ cooler!

Network (channels):

- ↳ Synchronous model: Have an upper-bound on speed (time to deliver a message).
- ↳ Asynchronous model: Don't have an upper-bound.

→ much cooler!

→ They can:

- ↳ Lose messages
- ↳ Create messages
- ↳ Duplicate messages
- ↳ Reorder messages

IP doesn't resolve these problems
TCP does!

Models

Fair-loss Models: NO duplication, NO creation, message sent ∞ times is received ∞ number of times (eventually we get the message), reordering $\leq$ FIFO

Reliable Channel Model: NO anything.

Processes

Can FAIL:

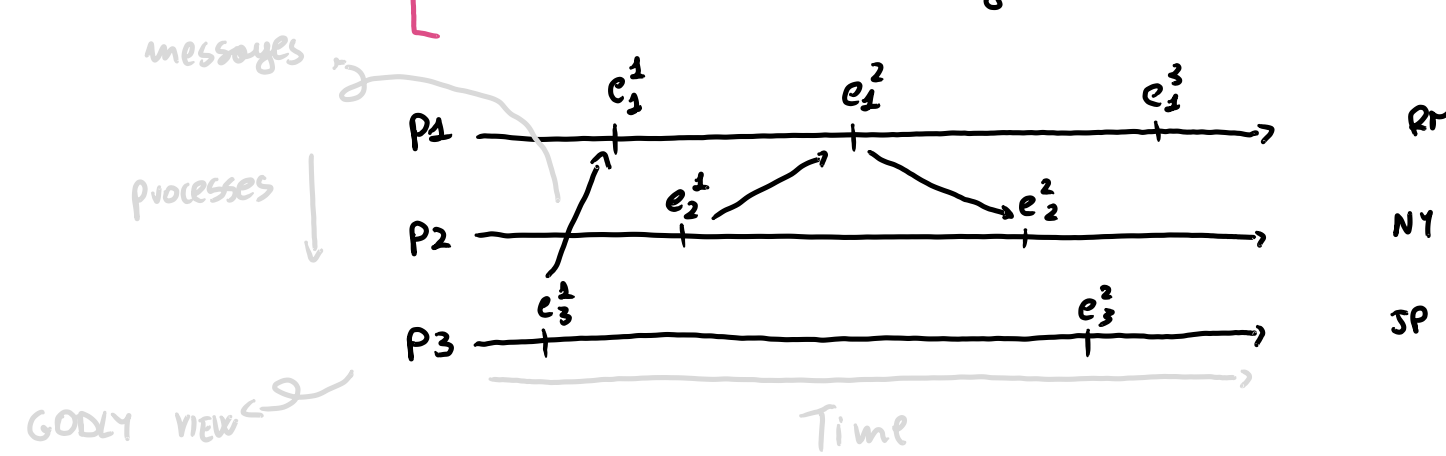
- ↳ Crash-stop: Stops.
- ↳ Crash-recover: Stops but self-healing is possible
- ↳ Byzantine: They do the fuck they want (also malicious things) → "two Byzantine General Problems"

The FLP (Fischer-Lynch-Paterson)

In an asynchronous deterministic system with just 1 crash-stop failure, the consensus problem is not solvable

N.B. In consensus we do both:

- Safety: We do the same thing (ex. we attack at the same time in the two general problem)
- Liveness: We do something



event number y in process x
↳ They can be anything (ex starting processes, receiving/sending messages)

↳ This is an asynchronous system so today we see this configuration, tomorrow's could be different.

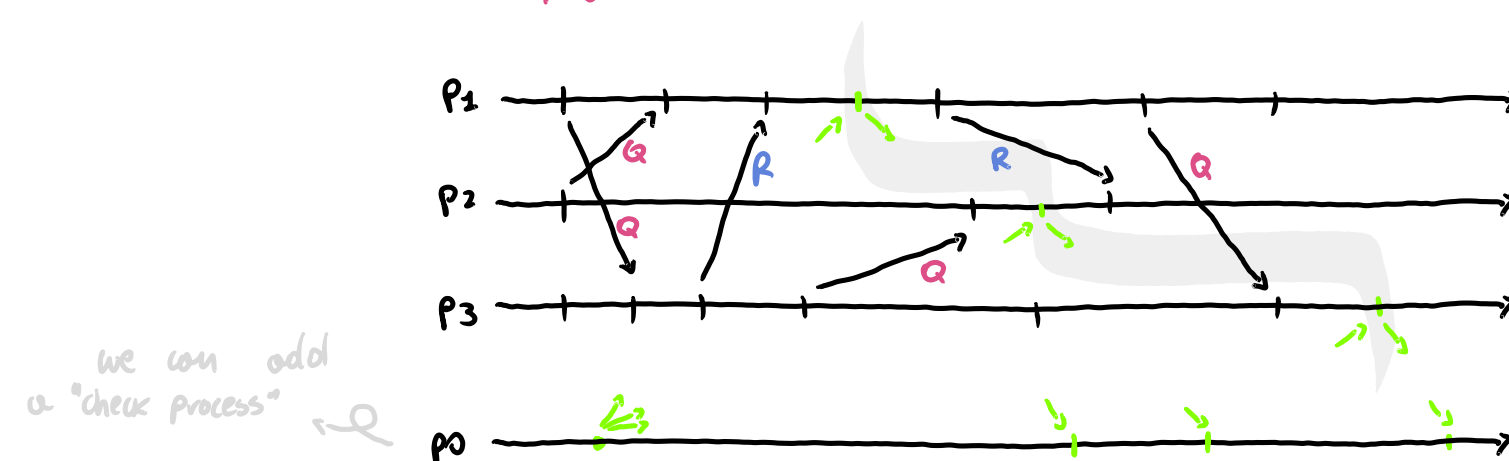
Definition: Local State (state of registers and other storages)

Signature σ_i^k → State of process i after $e_i^1 \dots e_i^k$ → "snapshot of the situation"

Definition: Global State

$\sigma^k = (\sigma_1^k, \dots, \sigma_m^k)$ → All local states

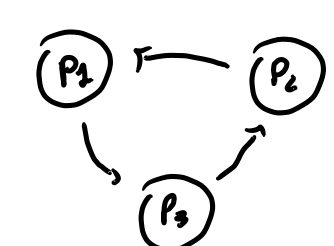
Example:



DEADLOCKS CAN HAPPEN!

↳ Literally saying to processes "Who are you waiting for?"

→ By observing the checks that we do we can determine this graph



$\boxed{\times} \Rightarrow \boxed{\gamma}$: x is waiting for y

→ That's a deadlock!

→ But looking at the execution we see no deadlock!

→ We are representing a global state that is not possible (variation in time in the different checks / "snapshot")
↳ This state cannot exist

continues...